



Al-Khwarizmi University Admission Test 2026

M.Sc. Applied Artificial Intelligence

Duration: **1 Hour**

You will need:

- Multiple choice answer sheet
- Soft, clean eraser
- Pencil

SAMPLE

1. If set $B = \{a, b, c, d\}$, how many elements are in the power set of B?

- A. 8
- B. 12
- C. 16
- D. 20
- E. None of the above

2. What is an "object" in object-oriented programming?

- A. A function that returns no value
- B. A blueprint used only at compile time
- C. An instance of a class containing data and behavior
- D. A loop that repeats indefinitely
- E. None of the above

3. A matrix has 5 rows and 2 columns. How many elements does it contain?

- A. 7
- B. 10
- C. 25
- D. 52
- E. None of the above

4. Which data structure uses nodes connected by pointers, where each node points to the next, allowing efficient insertion and deletion without shifting elements?

- A. Array
- B. Linked List
- C. Hash Table
- D. Stack
- E. None of the above

5. How many edges does a complete graph with 6 vertices (K_6) have?

- A. 12
- B. 15
- C. 18
- D. 30
- E. None of the above

6. What is the primary purpose of a destructor in a class?

- A. To initialize object data members
- B. To release resources and perform cleanup when an object is destroyed
- C. To create copies of objects
- D. To overload operators
- E. None of the above

7. A fair six-sided die is rolled twice. What is the probability that the sum of the two rolls equals 7?

- A. $1/12$
- B. $1/9$
- C. $1/6$
- D. $1/4$
- E. None of the above

8. What is the time complexity of inserting an element at the beginning of a singly linked list?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n^2)$
- E. None of the above

9. What is the output of the following code?

```
int a = 10;  
a = a - 4;  
cout << a;
```

- A. 4
- B. 6
- C. 10
- D. 14
- E. None of the above

10. What is a scalar, as opposed to a vector, in the linear algebra used in machine learning?

- A. A quantity with both magnitude and direction
- B. A single number with magnitude only, and no direction
- C. A table of numbers arranged in rows and columns
- D. A function that maps vectors to vectors
- E. None of the above

11. Which of the following is NOT one of the four pillars of object-oriented programming?

- A. Inheritance
- B. Encapsulation
- C. Compilation
- D. Polymorphism
- E. None of the above

12. What is the worst-case time complexity of searching for an element in an unsorted array of n elements?

- A. $O(1)$
- B. $O(\log n)$
- C. $O(n)$
- D. $O(n \log n)$
- E. None of the above

13. What is the probability of drawing an ace from a standard deck of 52 playing cards?

- A. $1/52$
- B. $1/13$
- C. $1/4$
- D. $1/26$
- E. None of the above

14. Which traversal of a binary tree visits the root node before its left and right subtrees?

- A. Inorder
- B. Postorder
- C. Preorder
- D. Level-order
- E. None of the above

15. Which OOP principle allows a single function name to behave differently depending on the object or arguments it is called with?

- A. Encapsulation
- B. Polymorphism
- C. Abstraction
- D. Recursion
- E. None of the above

16. A relation that is reflexive, transitive, and antisymmetric (but not necessarily symmetric) is called a:

- A. Equivalence relation
- B. Partial order
- C. Bijection
- D. Function
- E. None of the above

17. Matrix A has dimensions 3×2 and matrix B has dimensions 2×4 . What are the dimensions of the product $A \times B$?

- A. 3×4
- B. 2×2
- C. 4×3
- D. 3×2
- E. The multiplication is not defined

18. In C++, which access specifier makes class members accessible only from within the class itself?

- A. public
- B. private
- C. protected
- D. static
- E. None of the above

19. A binary tree has 4 levels, with the root counted as level 1. What is the maximum number of nodes it can contain?

- A. 8
- B. 15
- C. 16
- D. 31
- E. None of the above

20. What is the dot product of the vectors (2, 3) and (4, 1)?

- A. 8
- B. 9
- C. 11
- D. 14
- E. None of the above

21. What is the output of the following code?

```
int product = 1;
for (int i = 1; i <= 4; i++) {
    product *= i;
}
cout << product;
```

- A. 4
- B. 10
- C. 24
- D. 16
- E. None of the above

22. In logic, what is the result of the OR (V) operation when both inputs are false?

- A. True
- B. False
- C. Undefined
- D. It depends on context
- E. None of the above

23. What is the median of the data set {3, 7, 9, 12, 20}?

- A. 7
- B. 9
- C. 10.2
- D. 12
- E. None of the above

24. Which data structure is best suited for implementing an "undo" feature in a text editor?

- A. Queue
- B. Stack
- C. Binary Search Tree
- D. Hash Table
- E. None of the above

25. In how many distinct ways can the letters of the word "LEVEL" be arranged?

- A. 20
- B. 30
- C. 60
- D. 120
- E. None of the above